COHOMOLOGICAL CLASSIFICATION OF DEFINABLE GALOIS EXTENSIONS FOR FIXED INTERNALITY DATA VIA BINDING GROUPOIDS

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Allowed standard prerequisites (editorial).

- Totally transcendental theories eliminating imaginaries, internality, definable torsors and Galois cohomology. Original precise prime-model/parameter setup, complete binding-groupoid construction and author bibliography retained.

1. Introduction

In this paper we give two kinds of results which relate the definable Galois cohomology of [18] to the definable Galois theory of [27] in the setting of the prime model of a totally transcendental theory. Both sets of results draw inspiration from the theory of differentially closed fields of characteristic 0, where we look at interactions between the Picard-Vessiot theory and (constrained) differential Galois cohomology as presented in [7]. In this setting we give a new colimit formula for differential Galois cohomology and propose a new definition of boundedness for a differential field.

Our differential algebraic results continue a line of work [19] [20] [21] [16] [2] [14] giving fundamental triviality and boundedness results for Galois cohomology in the setting of a differential field K of characteristic 0. We begin by reviewing the basic facts in the algebraic setting which this work generalizes.

Boundedness is an important field arithmetic property which plays a role in existence results for Picard-Vessiot extensions over differential fields with non-algebraically closed fields of constants [6]. We recall the definition: A field F is bounded if it has finitely many Galois extensions of each finite degree.

Galois cohomology is a fundamental tool in algebraic number theory [25]. Let G be an algebraic group defined over a perfect field F. Then usual algebraic Galois cohomology $\mathrm{H}^1_{\mathrm{alg}}(F, G)$ is a pointed set classifying F-isomorphism classes of algebraic torsors for G. The basepoint corresponds to the class of G considered as a torsor.

There are a pair of fundamental relationships between torsors and Galois extensions which provide intuition: Firstly, a Galois extension of a field F gives rise to a torsor for the Galois group. Secondly, any torsor for an algebraic group becomes isomorphic to the group acting on itself over some finite Galois extension.

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The close connection between torsors and Galois extensions tells us first that Galois cohomology can be understood as a limit of finite extensions:

Fact 1.1. [25] *Let F be a perfect field and let G be an algebraic group defined over F. Then*

$$\varinjlim \mathrm{H}^1_{\mathrm{alg}}(E/F, G(E)) \cong \mathrm{H}^1_{\mathrm{alg}}(F, G)$$

where the limit is taken over the directed system of finite Galois extensions E of F and $\mathrm{H}^1_{\mathrm{alg}}(E/F, G(E))$ denotes the isomorphism classes of G-torsors which have an E-point.

It also tells us that algebraic closure can be detected cohomologically:

Fact 1.2. [25]*A perfect field F is algebraically closed if and only if $\mathrm{H}^1_{\mathrm{alg}}(F, G) \cong 1$ for all linear algebraic groups G defined over F.*

Lastly, it tells us that boundeness can be characterized cohomologically:

Fact 1.3. [25] *Let F be a perfect field. The following are equivalent*

1. *F is bounded.*
2. *For any linear algebraic group G defined over F, $\mathrm{H}^1_{\mathrm{alg}}(F, G)$ is finite.*
3. *For any linear algebraic group G defined over F there is a finite Galois extension E over F such that $\mathrm{H}^1_{\mathrm{alg}}(E/F, G(E)) \cong \mathrm{H}^1_{\mathrm{alg}}(F, G)$.*

In the final two sections of the paper we give versions of these statements in the setting of differential fields of characteristic 0 and in the setting of totally transcendental first order theories. To state our results we first review the conventions from [15] for the Picard-Vessiot theory and say a little bit about differential Galois cohomology.

Let K be an ordinary differential field of characteristic 0 with constant field C_K. In any differential field extension L of K, the solutions to a homogeneous linear differential equation $\mathcal{L}(y) = y^{(n)} + a_{n-1}y^{(n)} + ... + a_0 y = 0$ with $a_i \in K$, form a C_L-vector space of dimension n at most in L. A fundamental system of solutions to an equation $\mathcal{L}(y) = 0$ is a set of n-independent solutions over the constant field of any differential field containing them. A Picard-Vessiot extension of finite type is an extension L/K generated by a fundamental system of solutions to a homogeneous linear equation over K such that $C_L = C_K$. A Picard-Vessiot extension is a possibly infinite union (really, compositum in the differential closure K^{diff}) of Picard-Vessiot extensions of finite type.

The constrained differential Galois cohomology was introduced in the differential setting by Kolchin and Kovacic after a period of subsequent refinements [7] and is equivalent to the definable Galois cohomology specialized to the theory of differentially closed fields of characteristic 0, DCF_0. This cohomology theory measures isomorphism classes of differential algebraic torsors for differential algebraic groups (equivalently definable groups in DCF_0) as exposited in [7]. We review definable Galois cohomology below in Section 4.

Indeed versions of the basic relationships between Galois extensions and torsors from the algebraic setting extend to the differential setting. This takes the form of the well known "torsor theorem" from Picard-Vessiot theory [28]. Namely, for a homogeneous linear differential equation $\mathcal{L}(y) = 0$ over a differential field K the set of (realizations of a type of) fundamental systems of solutions in K^{diff} over $K(C_{K^{\mathrm{diff}}})$ form a right torsor for the Galois group attached to the extension

(generated by any realization). This theorem and it's converse, namely that any Picard-Vessiot extension can be generated by a suitable equation on a certain torsor, were generalized to the totally transcendental setting in [26]. We quote the result as Theorem 6.7 at the end of Section 6.

The analogue of the algebraic triviality result, Fact 9.1 above, in the differential setting is from [21]. This theorem was recently generalized to the setting of difference Picard-Vessiot theory in [1] and earlier partially to the multiple derivation setting in [2]. A version of the same result also appears in [16]:

Theorem 1.4. [21] *Let K be a differential field of characteristic 0. Then K is algebraically closed and closed under Picard-Vessiot extensions if and only if for any linear differential algebraic group G over K, $\mathrm{H}^1_\delta(K, G)$ is trivial.*

Our main contribution to this line of work in the differential field setting are differential versions of the algebraic colimit formula of Fact 9.2. A version of this fact for constrained cohomology appears in [7] but we find it unsatisfactory because the colimit is taken essentially over the family of all finitely differentially generated extensions of K in K^{diff}.

Before stating the analogue of Fact 9.2, we point out two well known differences between the algebraic and differential algebraic settings. Firstly, the union of all Picard-Vessiot extensions $(K)^{PV} = K^{PV_1}$ of K in K^{diff} is not the differential closure, K^{diff}, and indeed is not even PV-closed. One must iterate this operation to obtain a sequence of PV-extensions

$$K \subseteq K^{PV} = K^{PV_1} \subseteq (K^{PV_1})^{PV} = K^{PV_2} \subseteq ...$$

whose union K^{PV_∞} is closed under PV extensions and is most of the time still not equal to K^{diff}. The theorem above, Theorem 1.4, can then be reformulated as in [14] saying that any torsor for a linear differential algebraic group over K has a K^{PV_∞}-point:

Theorem 1.5. [14] *Let K be a differential field of characteristic 0. Then for any linear differential algebraic group G over K, $\mathrm{H}^1_\delta(K^{PV_\infty}/K, G(K^{PV_\infty})) \cong \mathrm{H}^1_\delta(K, G)$.*

A second difference between the two settings is that the differential closure of a differential field may properly embed into itself over K. We let $\mathrm{mcl}(K)$ denote the intersection of all self embeddings breaking from notation in [15]. We then have a sequence of differential extensions of K all of which generalize, in different senses, the algebraic closure of a field:

$$K \subseteq K^{PV_1} \subseteq K^{PV_2} \subseteq ... \subseteq K^{PV_\infty} \subseteq \mathrm{mcl}(K) \subseteq K^{\mathrm{diff}}$$

Our first (and essentially tautological) observation is that:

Proposition 1.6. *Let G be a linear differential algebraic group defined over K with $C_K = (C_K)^{\mathrm{alg}}$. Then*

$$\varinjlim H^1_{\mathrm{def}}(L/K, G(L)) \cong H^1_{\mathrm{def}}(K^{PV_1}/K, G(K^{PV_1}))$$

where the limit is again taken over the family of Picard-Vessiot extensions of finite type.

In [10] and [15] a Galois correspondence for K^{PV_∞} is introduced and refined. It relates normal closures of iterated Picard-Vessiot extensions of finite type to relatively type definable subgroups of $\mathrm{Aut}(K^{PV_\infty}/K)$. In Section 9, using Theorem

1.5 together with results of [10] and [15] we give the following colimit description of $H^1(K, G)$ for a linear differential algebraic group G over K. Namely, we give a differential analogue of Fact 9.2:

Theorem 1.7. *Let G be a linear differential algebraic group defined over a differential field K with $C_K = (C_K)^{\mathrm{alg}}$. Then*

$$\varinjlim \mathrm{H}^1_\delta(L/K, G(L)) \cong \mathrm{H}^1_\delta(K^{PV_\infty}/K, G(K^{PV_\infty}))$$

where the limit is taken over the directed system of normal closures of iterated PV extensions of K of finite type. Note $\mathrm{H}^1_\delta(K^{PV_\infty}/K, G(K^{PV_\infty})) \cong \mathrm{H}^1_\delta(K, G)$ by Theorem 1.5.

At the end of Section 9 we then propose the following definition of differential boundedness (giving a definitional form of Fact 1.3) adding to a list of possible such definitions from [14]:

Definition 1.8. A differential field K with $C_K = (C_K)^{\mathrm{alg}}$ is said to be bounded with respect to a differential algebraic group G defined over K if there is an iterated PV extension of finite type, whose normal closure L, satisfies

$$\mathrm{H}^1_{\mathrm{def}}(L/K, G(L)) \cong \mathrm{H}^1_{\mathrm{def}}(K, G).$$

We now discuss generalizations of our colimit and boundedness results in the totally transcendental setting which we present in Section 10.

In [23], Poizat gave the first model theoretic treatment of the Picard-Vessiot theory using the machinery of the binding group associated to internality data. This came after an investigation of the structure and minimality properties of prime models of totally transcendental first order theories [22]. The differential closure K^{diff} of a differential field being an example of the prime model of a totally transcendental theory, in this case DCF_0. In [22], the minimal closure of a set of parameters A inside a copy of the prime model M is defined as the intersection of all A-elementary embeddings M into itself. We denote this $\mathrm{mcl}(A)$. The key property for Galois theory which $\mathrm{mcl}(A)$ enjoys is that any subset B such that $A \subseteq B \subseteq \mathrm{mcl}(A) \subseteq M$, $\mathrm{dcl}(B)$ is exactly the fixed points of the action of $\mathrm{Aut}(M/B)$ on M. Equivalently, M is strongly homogeneous over any such B. In [15], a Galois correspondence between dcl-closed subsets of $\mathrm{mcl}(A)$ and relatively type definable subgroups of $\mathrm{Aut}(M/A)$ is shown which in the case of differential fields gives the correspondence of [10]. In Section 3 we review the the minimal closure $\mathrm{mcl}(A)$ inside of the prime model M over a set of parameters A in a totally transcendental theory.

The first result of Section 10 is the following lemma which appears as Lemma 10.2:

Lemma 1.9. *Let G be an A-definable group. Let P be an A-definable PHS for G. If $G(M) = G(\mathrm{mcl}(A))$ then $P(M) = P(\mathrm{mcl}(A))$.*

For our notion of definable Galois extension we follow [27]. We review this in Section 6. Namely over a set of parameters A, with respect to a definable set X, a definable Galois extension $B = dcl(A, b)$ for b a realization of a type strongly internal and weakly orthogonal to X.

In Section 10, we define, following [15], an iterated definable Galois extension $A \subseteq B \subseteq M$ relative to X is a dcl-closed subset such that either there exists an infinite sequence of dcl-closed subsets $A = B_0 \subseteq B_1 \subseteq \cdots$ with each B_i a proper definable Galois extension of B_{i-1} such that $B = \cup_i B_i$, or there exists a finite such

sequence $A = B_0 \subseteq B_1 \subseteq \cdots B_n = B$. If B is the union of a finite such chain where each B_i/B_{i-1} is a definable Galois extension of finite type, we call B an iterated definable Galois extension of finite type of A relative to X. Similarly we define the sequence of definable Galois closures $A \subseteq A^{XDG_1} \subseteq A^{XDG_2} \subseteq ... \subseteq A^{XDG_\infty}$ to be the sequence of definable Galois closures with reference to some definable set X over A in analogy with the sequence of Picard-Vessiot closures above.

Using the Lemma 10.2 we prove the following:

Theorem 1.10. *Let G be an A-definable group with $G(M) = G(\mathrm{mcl}(A))$. Then*

(1) *The definable Galois cohomology of G over $\mathrm{mcl}(A)$ is trivial:*

$$\mathrm{H}^1_{\mathrm{def}}(M/\mathrm{mcl}(A), G(M)) = 1.$$

(2) *Using the short exact sequence*

$$\mathrm{H}^1_{\mathrm{def}}(\mathrm{mcl}(A)/A, G(\mathrm{mcl}(A))) \cong \mathrm{H}^1_{\mathrm{def}}(A, G)$$

Namely, all of the definable Galois cohomological information for G in M is contained in $\mathrm{mcl}(A)$.

(3) *Any definable Galois extension relative to G is contained in $\mathrm{mcl}(A)$.*

By the last part we obtain:

Proposition 1.11. *Let A be a dcl-closed set of parameters. Let X be an A-definable set such that $X(M) = X(\mathrm{mcl}(A))$. Then each A^{XDG_n} is contained in $\mathrm{mcl}(A)$ and we have*

$$A \subseteq A^{XDG_1} \subseteq A^{XDG_2} \cdots \subseteq A^{XDG_\infty} \subseteq \mathrm{mcl}(A) \subseteq M.$$

Furthermore each A^{XDG_n} is normal.

In Section 4 we recall the definition of definable Galois cohomology from [18]. For an A-definable group G in a totally transcendental theory eliminating imaginaries, the definable Galois cohomology classifies A-definable right principal homogeneous spaces (PHSs) or torsors for G. We use the short exact sequence of [13] to prove the following general lemma:

Lemma 1.12. *Let M be a copy of the prime model over A. Let B be normal in M over A. Let $\{B_i\}_{i\in I}$ be a family of normal subsets of B which form a directed system under inclusion and such that $\cup_{i\in I} B_i = B$. Then $\varprojlim Aut(B_i/A) \cong Aut(B/A)$ and moreover if G is an A-definable group then*

$$\varinjlim \mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_i)) \cong \mathrm{H}^1_{\mathrm{def}}(B/A, G(B)).$$

We then obtain the following colimit formula in the totally transcendental setting, the condition $X(M) = X(A)$ analogous to the assumption that the constants be algebraically closed in the differential setting:

Theorem 1.13. *Let G be an A definable group living on an A-definable set X with $X(M) = X(A)$. Then*

$$\varinjlim \mathrm{H}^1_{\mathrm{def}}(B/A, G(B)) \cong \mathrm{H}^1_{\mathrm{def}}(A^{XDG_\infty}/A, G(A^{XDG_\infty}))$$

where the limit is taken over the family of normal closures of iterated definable Galois extensions of finite type B of A relative to X.

In contrast with the differential case we do not have a colimit formula for the full extension M/A. We leave to future work an investigation of the extensions $A^{XDG_\infty} \subseteq \mathrm{mcl}(A)$ for various sets X. Nevertheless we feel the results in this section, together with the correspondence of [15], to be a meaningful response to the call in [22] for an investigation of the automorphism group of $\mathrm{mcl}(A)$ over A.

At the end of Section 10, we propose a definition of boundedness for a set of parameters:

Definition 1.14. Let X be an A-definable set satisfying $X(A) = X(M)$. Let G be an A-definable group living on X. The set of parameters A is said to be bounded with respect to G and X if there exists an iterated definable Galois extension of finite type relative to X whose normal closure B satisfies

$$\mathrm{H}^1_{\mathrm{def}}(B/A, G(B)) \cong \mathrm{H}^1_{\mathrm{def}}(A^{XDG_\infty}/A, G(A^{XDG_\infty})).$$

In Section 8 we give the other main result of the paper which is a generalization of the following result relating the multiplicity of Picard-Vessiot extensions to the algebraic Galois cohomology of the Galois group. There is no assumption here that the constant field be algebraically closed:

Theorem 1.15. [6][3] *Let $\mathcal{L}(y) = 0$ be an homogeneous linear ordinary differential equation (LODE) over a differential field K of characteristic 0. Assume that there is at least one nontrivial Picard-Vessiot extension L/K for $\mathcal{L}(y) = 0$ in K^{diff}, a differential closure of K. Let $G(C_{K^{\mathrm{diff}}})$ be the Picard-Vessiot differential Galois group of the extension L/K, a linear algebraic group defined over C_K. Then the usual algebraic Galois cohomology*

$$\mathrm{H}^1_{\mathrm{alg}}(C_K, G)$$

classifies (it is a pointed set whose cardinality counts) exactly the set of distinct Picard-Vessiot extensions for $\mathcal{L}(y) = 0$ in K^{diff}. The base point of the cohomology set corresponds to the extension L/K.

The generalization of Theorem 1.15 shown in Section 8 is the following:

Theorem 1.16. *Let T be a totally transcendental theory eliminating imaginaries. Let A be a* dcl*-closed set of parameters and M be a copy of the prime model over A. Let Y and X be A-definable sets with Y internal to X. Produce $f : Z \times X_1 \to Z$ as in the construction of the binding groupoid (given in detail in Section 7) where Z is the A-definable set of "fundamental systems" associated to Y, X_1 is an A-definable set contained in $\mathrm{dcl}^{eq}(A, X)$, and f is an A-definable function such that for any $b_1, b_2 \in Z$, there is a unique $c \in X_1$ with $f(b_1, c) = b_2$. Then assuming*

1. *there is at least one $b_0 \in Z(M) \backslash Z(A)$ with $\mathrm{dcl}(A, b_0) \cap \mathrm{dcl}(A, X) = A$ so that $A \subsetneq \mathrm{dcl}(A, b_0)$ is a proper definable Galois extension in the sense of* [27] *with extrinsic definable Galois group G_0,*
2. *the A-definable set X_1 carries an A-definable group structure such that Z is exactly an A-definable right PHS for X_1 with action f,*
3. *and $\mathrm{H}^1_{\mathrm{def}}(M/A, X_1(M))$ is trivial*

we have that $\mathrm{H}^1_{\mathrm{def}}(M/A, G_0(M))$ classifies exactly the set of definable Galois extensions in M for the internality data (Y, X). The base point of the cohomology set corresponds exactly to the extension $A \subsetneq \mathrm{dcl}(A, b_0)$.

In Section 8 we comment in more detail on the various assumptions. It is stated in a slightly more general form where we also consider the case $\mathrm{H}^1_{\mathrm{def}}(M/A, X_1(M))$ is nontrivial i.e., assumption 3) is dropped. In general, there seems to be no reason why assumption 2) needs to hold, i.e., X_1 needs to carry a group structure making (Z, X_1) into an A-definable PHS with action f. However in practice this very often is the case. Moreover, from the perspective of a fixed extension, this can always be taken to be the case as can be seen by the main result of [26].

The forward direction this theorem in the setting of DCF_0, i.e. that under suitable hypotheses $\mathrm{H}^1_{\mathrm{def}}(M/A, G_0(M))$ bounds the number of definable Galois extensions (this direction not needing assumption 3), is used in the existence results for strongly normal extensions of [6]. Much work has been done to extend these results to $DCF_{0,m}$ [9], [24] [8], but the actual extension, even just with respect to the constants, is not fully complete at the time of this paper in the strongly normal setting of $DCF_{0,m}$. We hope that the above theorem will find use extending these existence results to the generalized strongly normal theory, but the result should be of independent interest.

One can also see by inspection that the forward direction follows from the construction of the binding groupoid which can be seen as a generalization of the "torsor theorem" of differential Galois theory which we give a detailed account of in Section 7. The reverse direction is a straightforward application of the long exact sequence in cohomology developed in [8]. The proof here is an adaptation the proof of Theorem 1.15 via definable Galois cohomology given in Chapter 3 of [12]. The original proof goes back to at least [3] and is shown via tannakian methods. That there should be a generalization of this fundamental Theorem 1.15 via the binding-groupoid should maybe be unsurprising given the close connection between the tannakian formalism and the model-theoretic notion of the binding groupoid associated to internality data [5], [6].

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3. Prime models, the minimal closure, and Galois correspondences

In this section we review and collate some key definitions and constructions.

Let T be a complete totally transcendental (t.t.) theory eliminating imaginaries. Let $\bar{M}$ be a monster model of T which we will assume to be κ-saturated and strongly κ-homogeneous for a suitably sized cardinal κ. Let A be a small set of parameters.

As T is t.t., the isolated types in $S_n(A)$ are dense and consequently constructible models M over A exist. By the back-and-forth argument of Resayrre, constructible models are unique up to A-isomorphism and strongly ω-homogeneous over A i.e. for tuples a and b from M with the same type over A, there is an automorphism

$\sigma \in \mathrm{Aut}(M/A)$ taking a to b. Constructible models are also atomic over A. Shelah proved the following characterization of constructibility: In a t.t. theory a model M over A is constructible if and only if it is atomic over A and has no uncountable A-indiscernible sequence. It is clear that a constructible model is prime over A, and as we said, in the t.t. setting constructible models exist. A prime model N over A must embed in a constructible model over A, it is then clear that N satisfies the criterion of Shelah's characterization and so is then itself constructible. The uniqueness atomicity, and strong ω-homogeneity of the prime model then follow. Alternatively one can show directly (without Shelah's characterization) that an arbitrary subset of a constructible model is still constructible in the t.t. setting.

Let M be a copy of the prime model over A of T. An intermediate extension $A \subseteq B \subseteq M$, is called normal if $\forall b \in B$ and $b' \in M$, $tp(b/A) = tp(b'/A)$ implies that $b' \in B$. By the strong ω-homogeneity of M over A, we have a short exact sequence of groups $1 \to \mathrm{Aut}(M/B) \to \mathrm{Aut}(M/A) \to \mathrm{Aut}(B/A) \to 1$. Finally, we remark that M remains prime over B.

Although the prime model over a set of parameters of a t.t. theory is unique up to isomorphism over that set, it is not necessarily minimal, i.e., it may properly elementarily embed into itself.

Definition 3.1. [22] Let M be a copy of the prime model over A. Recall, the minimal closure of A in M is the intersection $\cap_\gamma \gamma(M)$ where $\gamma : M \to M$ ranges over all elementary embeddings of M into itself. We denote this $\mathrm{mcl}(A)$ in analogy with the model theoretic definable and algebraic closures, denoted dcl and acl, respectively. A type $p \in S(A)$ is called atomically algebraic if it is isolated but has no infinite A-indiscernible set of realizations which is atomic over A. An element $a \in \bar{M}$ is atomically algebraic over A if $tp(a/A)$ is atomically algebraic.

Fact 3.2. [22] *Let M be prime over A in $\bar{M}$.*

1. $\mathrm{mcl}(A)$ *is equal to the set of atomically algebraic elements in M. Consequently, $M = \mathrm{mcl}(A)$, i.e. M is minimal, if and only if it contains no infinite atomic indiscernible sequence.*
2. $mcl(mcl(A)) = \mathrm{mcl}(A)$
3. *M remains constructible and hence prime over any subset of $mcl(A)$.*
4. *For any subsets B and B' of $mcl(A)$, if $tp(B/A) = tp(B'/A)$, there is an automorphism $\sigma \in Aut(M/A)$ taking B to B'.*

We also then have now three types of intermediate extensions $A \subseteq B \subseteq M$ over which the prime model remains prime, and hence atomic and strongly ω-homogeneous: Firstly, finitely dcl-generated extensions, i.e. when $B = \mathrm{dcl}(A, b)$ for a finite tuple b. Secondly for normal extensions B. Thirdly for arbitrary subsets B of $mcl(A)$. The use of this is principally in the Galois definability lemma in the context of a prime model (see [15] for instance):

Fact 3.3. [15] *Let $A \subseteq B \subseteq M$ be an intermediate extension over which M remains prime.*

1. *Let X be an M definable set and let $X(M)$ be $Aut(B/A)$-invariant. Then X is B-definable.*
2. *The fixed points of the action of $Aut(M/B)$ on M, denoted $M^{Aut(M/B)}$ is exactly* $\mathrm{dcl}(B)$.

This fact underpins both definable Galois theory and definable Galois cohomology in the setting of a prime model of a totally transcendental theory:

We need a few more definitions from [15].

Definition 3.4. [15] Let B be a dcl-closed intermediate set $A \subseteq B \subseteq \mathrm{mcl}(A) \subseteq M$. Let $\bar{b}$ be an enumeration of B. Let $p_0(\bar{x})$ be $tp(\bar{b}/A)$, and $S_{p_0}(B)$ the space of extensions of p_0 to B. Let X be a closed subset of $S_{p_0}(B)$ given by a set of formulas $\Sigma(\bar{x}, \bar{b})$ and Y be a clopen subset of $S_{p_0}(B)$ given by a formula $\varphi(x, b)$ for some b, a finite tuple from B.

(1) By $Y(\mathrm{Aut}(B/A))$ we mean a subset of the form
$$\{\sigma \in \mathrm{Aut}(B/A) : \models \varphi(b, \sigma(b))\}$$
which we call a relatively definable subset of $\mathrm{Aut}(B/A)$.

(2) By $X(\mathrm{Aut}(\bar{M}/A))$ we mean
$$\{\sigma \in \mathrm{Aut}(\bar{M}/A) : \models \Sigma(\bar{b}, \sigma(\bar{b}))\}$$
and we call this a relatively type definable over B subset of $\mathrm{Aut}(\bar{M}/A)$.

(3) By $X(\mathrm{Aut}(B/A))$ we mean $X(\mathrm{Aut}(\bar{M}/A))|_{\mathrm{Aut}(B/A)}$, i.e. a subset of $\mathrm{Aut}(B/A)$ of the form
$$\{\sigma \in \mathrm{Aut}(B/A) : \models \Sigma(\bar{b}, \sigma(\bar{b}))\}$$
which we call a relatively type definable over B subset of $\mathrm{Aut}(B/A)$.

A relatively (type) definable subgroup of $\mathrm{Aut}(B/A)$ is then a relatively (type) definable subgroup of $\mathrm{Aut}(B/A)$.

The following theorem generalizes the Galois correspondence for $\mathrm{acl}^{eq}(A)$ to $\mathrm{mcl}^{eq}(A)$.

Theorem 3.5. [15] *Let B be a* dcl*-closed normal intermediate set $A \subseteq B \subseteq \mathrm{mcl}(A) \subseteq M$. Then there is a Galois correspondence between dcl-closed subsets C of B containing A, and closed subgroups H of $Aut(B/A)$. Namely,*
$$C \mapsto H_C := \{\sigma \in Aut(B/A) : \ \sigma(c) = c \ \forall c \in C\}$$
and
$$H \mapsto B^H := \{b \in B : h(b) = b \ \forall h \in H\}.$$
Moreover, finitely dcl*-generated extensions correspond to relatively definable subgroups and normal intermediate extensions correspond to normal relatively type definable subgroups.*

4. Definable Galois cohomology and G-primitives

The only novelty in this section is the colimit formula in definable Galois cohomology.

Let M be a copy of the prime model over A and B a normal intermediate extension of parameters. Let G be an A-definable group.

Recall from [18], [13] that a map $\varphi : \mathrm{Aut}(B/A) \to G(B)$ is called an A-definable cocycle if it satisfies the usual cocycle condition, $\forall \sigma, \tau \in \mathrm{Aut}(B/A)$
$$\varphi(\sigma\tau) = \varphi(\sigma)\sigma(\varphi(\tau))$$
and there exists a tuple $b \in B$ and A-definable function $h(x, y)$ such that $\forall \sigma \in \mathrm{Aut}(B/A)$
$$h(b, \sigma(b)) = \varphi(\sigma).$$

Two cocycles φ and ψ are cohomologous if there exists an element $g \in G(B)$ such that $\forall \sigma \in \mathrm{Aut}(B/A)\ \varphi(\sigma) = g^{-1}\psi(\sigma)\sigma(g)$.

Definition 4.1. [13] The first definable Galois cohomology set $\mathrm{H}^1_{\mathrm{def}}(B/A, G(B))$ is defined to be the set of cohomology classes of definable cocycles $\varphi : \mathrm{Aut}(B/A) \to G(B)$. When $B = M$ we write $\mathrm{H}^1_{\mathrm{def}}(A, G)$ simply.

An A-definable right principal homogeneous space (PHS) or torsor for G is an A-definable set X with a strictly transitive A-definable action by G. Recall from [13] that $\mathrm{H}^1_{\mathrm{def}}(B/A, G(B))$ is isomorphic to (i.e. in basepoint preserving bijection with) the isomorphism classes of definable PHSs for G which have a B-point.

The main available tools for working with these cohomology sets are the following two theorems, Theorem 4.2 and 4.4 below:

Theorem 4.2. [13] *Let M be a copy of the prime model over A and $A \subseteq B \subseteq C \subseteq M$ be intermediate* dcl*-closed sets of parameters with B and C normal in M over A. Let G be an A-definable group. Then the short exact sequence of automorphism groups*

$$1 \to Aut(C/B) \to Aut(C/A) \to Aut(B/A) \to 1$$

induces a short exact sequence in definable Galois cohomology[1]

$$1 \to \mathrm{H}^1_{\mathrm{def}}(B/A, G(B)) \to \mathrm{H}^1_{\mathrm{def}}(C/A, G(C)) \to \mathrm{H}^1_{\mathrm{def}}(C/B, G(C)).$$

We obtain from this theorem the following general colimit formula for definable Galois cohomology. We will use this lemma in the final section for specific normal extensions B and directed systems.

Lemma 4.3. *Let B be normal in M over A. Let $\{B_i\}_{i\in I}$ be a family of normal subsets of B which form a directed system under inclusion and such that $\cup_{i\in I} B_i = B$. Then $\varprojlim Aut(B_i/A) \cong Aut(B/A)$ and moreover if G is an A-definable group then*

$$\varinjlim \mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_i)) \cong \mathrm{H}^1_{\mathrm{def}}(B/A, G(B)).$$

Proof. Note that $\varinjlim B_i$ is the disjoint union of the B_i modulo the relation of eventual equality. Each B_i includes into B which induces a map from $\varinjlim B_i$ to B. It follows immediately from the definitions that this is well defined and injective. Surjectivity follows from the assumption that $\cup_{i\in I} B_i = B$. So there is a canonical bijection from $\varinjlim B_i$ to B.

As $\{B_i\}_{i\in I}$ forms a directed system, $\{\mathrm{Aut}(B_i/A)\}_{i\in I}$ forms an inverse system. For each index i, the quotient homomorphism $Aut(B/A) \to \mathrm{Aut}(B_i/A)$ is surjective and the required compatibilities hold so that we have a map $Aut(B/A) \to \varprojlim \mathrm{Aut}(B_i/A)$. Injectivity follows by noting that if two automorphism of $Aut(B/A)$ differ, then they differ on some B_i. To check surjectivity, let $(\sigma_i)_{i\in I} \in \varprojlim \mathrm{Aut}(B_i/A)$. We define an automorphism of B whose image is $(\sigma_i)_{i\in I}$. For $b \in B$, there is an $i \in I$ such that $b \in B_i$, so define $\sigma(b) = \sigma_i(b)$. To see that this is well defined independent of the choice of B_i, let $i, j \in I$ such that $b \in B_i \cap B_j$. Then there is some B_k containing both B_i and B_j. Then $\sigma_i(b) = \pi_{ki}(\sigma_k)(b) = \sigma_k(b) = \pi_{kj}(\sigma_k)(b) = \sigma_j(b)$ where $\pi_{ki} : \mathrm{Aut}(B_k/A) \to \mathrm{Aut}(B_i/A)$ and $\pi_{kj} : \mathrm{Aut}(B_k/A) \to \mathrm{Aut}(B_j/A)$ are the projections. As σ restricts to an elementary permutation on each B_i, and $B = \cup_{i\in I} B_i$, σ is an elementary permutation. So $\mathrm{Aut}(B/A) \cong \varprojlim \mathrm{Aut}(B_i/A)$.

[1]Note that we ignore the transform action of $\mathrm{Aut}(B/A)$ on the final term. See Definition 4.1 of [13]

By Theorem 4.2, as the $\{\mathrm{Aut}(B_i/A)\}_{i\in I}$ form an inverse system of groups, $\{\mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_i))\}_{i\in I}$ forms a directed system of pointed sets. The inclusion of each B_i in B induces again via Theorem 4.2 a map from $\varinjlim \mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_i))$ to $\mathrm{H}^1_{\mathrm{def}}(B/A, G(B))$. As $\cup_{i\in I} B_i = B$, this map is surjective as the definability data for any cocycle in $\mathrm{H}^1_{\mathrm{def}}(B/A, G(B))$ must be contained in from some B_i. Injectivity is immediate: If for some indices $i, j \in I$ there are class in $\mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_i))$ and $\mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_j))$ which have the same image in $\mathrm{H}^1_{\mathrm{def}}(B/A, G(B))$, then they must be equal in $\mathrm{H}^1_{\mathrm{def}}(B_k/A, G(B_k))$ where $B_i, B_j \subseteq B_k$. So $\varinjlim \mathrm{H}^1_{\mathrm{def}}(B_i/A, G(B_i)) \cong \mathrm{H}^1_{\mathrm{def}}(B/A, G(B))$.

□

The following theorem of [8] is the other fundamental method for computing with Galois cohomology in the definable setting.

Theorem 4.4. [8] *Let H be an A-definable subgroup of a definable group G. We then have an exact sequence of pointed sets*

$$1 \to H(A) \to G(A) \to (G/H)(A) \xrightarrow{\delta} \mathrm{H}^1_{\mathrm{def}}(A, H) \to \mathrm{H}^1_{\mathrm{def}}(A, G)$$

where $(G/H)(A)$ is the pointed set of A-definable left cosets of H in G.

The connecting homomorphism δ of the above theorem identifies the A-definable left cosets of H in G among all right A-definable PHSs for H. The primitive elements in case (1) of the following definition are exactly basepoints of A-definable PHSs for H which are in the image of δ. The primitive elements of case (2) (for nontrivial W) are basepoints for A-definable PHSs for H which lie outside of the image of δ.

Definition 4.5. Let H be an A-definable subgroup of a definable group G.

(1) [17] Let $\pi : G \to G/H$ be the A-definable quotient map given by elimination of imaginaries. An element $a \in G$ is said to be a (G, H)-primitive over A if $\pi(a) \in (G/H)(A)$.
(2) [12] Let W be an A-definable right torsor for G and $\pi : W \to W/H$ the A-definable quotient map given by elimination of imaginaries. An element $a \in W$ is said to be a (W, G, H)-primitive over A if $\pi(a) \in (W/H)(A)$.

5. Internality

The material in this section is a standard part of geometric stability theory but we include it to be able to carefully state and prove the results of Section 8 as conventions vary by author.

Let T be a totally transcendental theory eliminating imaginaries. Let $\bar{M}$ be a monster model of T. Assume the language has at least two constants. Let A be a small dcl-closed set of parameters in $\bar{M}$. Let M be the prime model of T over A.

Definition 5.1. Let Y and X be a A-definable sets. Let $\psi_Y(\bar{z})$ and $\varphi_X(\bar{w})$ be A-definable formulas defining Y and X respectively. We say that Y is internal to X if there is a small subset of parameters $B \subset \bar{M}$ such that $Y \subseteq \mathrm{dcl}(B, X, A)$.

We set up notation:

Remark 5.2. Y is internal to X means that, for each $\bar{a} \in Y$, there is an A-formula, $\varphi_{\bar{a}}(\bar{y}_{\bar{a}}, \bar{x}_{\bar{a}}, \bar{z})$, a tuple $\bar{b}_{\bar{a}} \in B^{|\bar{y}_{\bar{a}}|}$ (the variable lengths can change depending on the

$\bar{a}$) and a tuple $\bar{c}_{\bar{a}} \in X^{|\bar{x}_a|}$ such that $\models \varphi_{\bar{a}}(\bar{b}_{\bar{a}}, \bar{c}_{\bar{a}}, \bar{a})$ and $\bar{a}$ is the unique element of $\bar{M}$ satisfying the above formula.

Lemma 5.3. *Let Y be X-internal as above. Then there exists an A-definable function $f(\bar{y}, \bar{x})$ and a tuple of elements $\bar{b}$ from Y such that for each $\bar{a} \in Y$, there exists a tuple $\bar{c}$ from $dcl(X, A)$ such that $f(\bar{b}, \bar{c}) = \bar{a}$.*

Remark 5.4. The tuple $\bar{c}$ in most cases will come from X. In the case that X and A are both empty, the assumption that the language has multiple constants guarantees that $dcl(X, A)$ is still at least countably infinite: Note that if there are at least two constants in the language then finite binary sequences may be coded.

Proof. Note that A and B are small. By assumption, the collection of $A \cup B$-formulas

$$\{\psi_Y(\bar{z})\} \cup \{\neg\exists \bar{x}_{\bar{a}} \in X^{|\bar{x}_{\bar{a}}|}(\varphi_{\bar{a}}(\bar{b}, \bar{x}_{\bar{a}}, \bar{z})) : \bar{b} \in B^{\bar{y}_{\bar{a}}}\}_{\bar{a} \in Y}$$

is inconsistent. Therefore it is finitely inconsistent by compactness. Thus there are finitely many $\bar{a}_1, ..., \bar{a}_m \in Y$ such that

$$\models \forall \bar{z} \left(\psi_Y(\bar{z}) \rightarrow \bigvee_{i=1}^{m} \exists \bar{x}_{\bar{a}_i} \in X^{|\bar{x}_{\bar{a}_i}|} \ \varphi_{\bar{a}_i}(\bar{b}_{\bar{a}_i}, \bar{x}_{\bar{a}_i}, \bar{z}) \right).$$

Let $e_1, ..., e_m$ be distinct elements from $dcl(X, A)$ (which exist by the preceding remark) associated to variables $u_1, ..., u_m$. Then for each $\bar{a} \in Y$, there exists $\bar{c}_{\bar{a}} \in X^{|\bar{x}_{\bar{a}_i}|}$ and $e_j \in dcl(X, A)$ such that

$$\models \bigvee_{i=1}^{m} (\varphi_{\bar{a}_i}(\bar{b}_{\bar{a}_i}, \bar{c}_{\bar{a}}, \bar{a}) \wedge e_i = e_j)$$

and $\bar{a}$ is the only such solution.

Denote the above formula by $\varphi(\bar{y}, \bar{x}, \bar{z})$ where we let $\bar{b} = (b_{\bar{a}_1}, ..., b_{\bar{a}_m})$ in the variables $\bar{y} = (\bar{y}_{\bar{a}_1}, ..., \bar{y}_{\bar{a}_m})$ and $\bar{x} = (\bar{x}_{\bar{a}_1}, u_1, ..., \bar{x}_{\bar{a}_m}, u_m, u)$ with u being the variable filled by the element e_j. It is clear that $\varphi(\bar{b}, \bar{x}, \bar{z})$ defines the graph of a definable function.

Let $\bar{a} \in Y$ now and $\bar{c} \in dcl(X, A)$ such that $\models \varphi(\bar{b}, \bar{c}, \bar{a})$. Then $\varphi(\bar{y}, \bar{c}, \bar{a}) \in tp(\bar{b}/dcl(X, A) \cup Y)$. As T is totally transcendental, by definability of types over arbitrary subsets of $\bar{M}$, $\varphi(\bar{b}, \bar{x}, \bar{z})$ is definable over $dcl(X, A) \cup Y$.

We now make some replacements in notation and we are finished.

The set defined by $\varphi(\bar{b}, \bar{x}, \bar{z})$, is defined by a $dcl(X, A) \cup Y$-formula which we will now write as $\varphi_0(\bar{b}, \bar{c}, \bar{x}, \bar{z})$. Where $\bar{b} = (b_1, ..., b_n)$ is a new tuple of parameters from Y, $\bar{c}$, in say variables $\bar{v}$, is a new tuple of parameters from $\mathrm{dcl}(A, X)$ and $\varphi_0(\bar{y}, \bar{v}, \bar{x}, \bar{z})$ is A-definable. As before, $\varphi_0(\bar{b}, \bar{c}, \bar{x}, \bar{z})$ is the graph of a definable function. Now both $\bar{v}$ and $\bar{x}$ are sorts on $\mathrm{dcl}(A, X)$ so rewrite them as just $\bar{x}$. Then $\varphi_0(\bar{b}, \bar{x}, \bar{z})$ is still clearly a the graph of a definable function $f(\bar{b}, \bar{x}) = \bar{z}$.

The tuple $\bar{x} = (\bar{x}_{\bar{a}_1}, u_1, ..., \bar{x}_{\bar{a}_m}, u_m, u, \bar{v})$. So the $\bar{x}$ variable ranges over an A-definable subset X_0 of $\mathrm{dcl}(A, X)$. □

Definition 5.5. We call the tuple $\bar{b} = (b_1, ..., b_n)$ a fundamental system of solutions associated to Y, X.

Remark 5.6. Tracing through the above proof we can see that if X is the empty set, then Y is algebraic over a finite extension of A from B i.e.

$$\models \forall \bar{z} \left(\varphi_Y(\bar{z}) \to \bigvee_{i=1}^{m} \varphi_{\bar{a}_i}(\bar{b}_{\bar{a}_i}, \bar{z}) \right)$$

with the disjunction being an algebraic formula. That is, Y implies an algebraic formula over $\mathrm{dcl}(A, \bar{b}_{\bar{a}_1}, ..., \bar{b}_{\bar{a}_m})$. Conversely, any algebraic A-definable set Y is internal to the $\emptyset$, because it is in the definable closure of A union a full (finite) set of solutions to Y in $\bar{M}$ which is finite and therefore small. That is, for any small A-definable set Y (which then must be algebraic), $Y \subseteq \mathrm{dcl}(A, Y)$ witnesses internality.

Lemma 5.7. *Let Y be internal to X, both defined over a small* dcl*-closed set A. The set of fundamental systems $Z \subset Y^n$ is A-definable.*

Proof. The A-definable formula

$$\forall \bar{z} \in Y \exists \bar{x} \in X_0 \ f(\bar{y}, \bar{x}) = \bar{z}$$

in the variable $\bar{y}$, defines the subset $Z \subseteq Y^n$. Note that this n is the length of any fundamental system i.e., the n in the proof of Lemma 5.3. □

Remark 5.8. As M is an elementary substructure of $\bar{M}$, the previous lemma shows that we can find a fundamental system $\bar{b}$ in M, the prime model over A.

Lemma 5.9. *Let Y and X be A-definable with Y internal to X. Let Z be the associated A-definable set of fundamental systems. Then there is an A-definable subset X_1 of $dcl(A, X)$ and an A-definable function $f(\bar{y}, \bar{x}) : Z \times X_1 \to Z$ such that for any $\bar{b}, \bar{b}' \in Z$ there is a unique $\bar{c} \in X_1$ with $f(\bar{b}, \bar{c}) = \bar{b}'$.*

Proof. Let $b \in Y$. For any $\bar{b} \in Z \subseteq Y^n$ there is a $\bar{c} \in X_0$ such that $f(\bar{b}, \bar{c}) = b$.

Then for any $\bar{b}, \bar{b}' = (b'_1, ..., b'_n) \in Z$ there is a tuple $(\bar{c}_1, ..., \bar{c}_n) \in (X_0)^n$ such that $f(\bar{b}, \bar{c}_i) = b'_i$.

Define an equivalence relation on $(X_0)^n$ by saying $\hat{c} = (\bar{c}_1, ..., \bar{c}_n)$ and $\hat{c}' = (\bar{c}'_1, ..., \bar{c}'_n)$ are equivalent if

$$\forall \bar{y} \in Z \ \bigwedge_{i=1}^{n} f(\bar{y}, \bar{c}_i) = f(\bar{y}, \bar{c}'_i).$$

The quotient of $(X_0)^n$ by this A-definable equivalence relation gives another A-definable set X_1 living in $\mathrm{dcl}^{\mathrm{eq}}(X, A)$ which by EI is just $\mathrm{dcl}(X, A)$. We then replace $f(\bar{y}, \bar{x})$ in notation with $f(\bar{y}, \bar{x}_1), ..., f(\bar{y}, \bar{x}_n)$ with the $\bar{x}_i$ ranging over X_1. With $f(\bar{y}, \bar{x}_i)$ being the induced function $Z \times X_1 \to Y$.

Now $f(\bar{y}, \bar{x})$ is then an A-definable function function $Z \times X_1 \to Y^n$. Lastly, by Lemma 5.7 we replace X_1 in notation with the definable subset of itself so that the image of this map is exactly Z. □

Example 5.10. Let Y be the set defined by

$$y^{(n)} + a_{n-1} y^{(n-1)} + ... + a_0 y = 0,$$

an ordinary homogeneous linear differential equation in scalar form defined over a differential field K.

The K-definable set Y is internal to the constants as its solutions in K^{diff} form an n-dimensional vector space over the constants $C_{K^{\mathrm{diff}}}$. The set of such fundamental

systems for Y that is, the set of bases for the vector space of solutions, is a K-definable set $Z(K^{\text{diff}})$.

The set $Z(K^{\text{diff}})$ is interdefinable with the set of fundamental matrices for Y in K^{diff}. We replace $Z(K^{\text{diff}})$ by this set of matrices. It will be a right PHS for the group $GL_n(C_{K^{\text{diff}}})$ which is X_1 in this example.

6. Definable Galois extensions

The notion of definable Galois extension which we use is the notion given in [27]. The only addition to the exposition here is we describe some properties of these extensions with respect to the minimal closure of a set of parameters inside of a prime model.

Let T be a complete totally transcendental theory eliminating imaginaries. Let $\bar{M}$ be a monster model. Let A be a small set of parameters. Let X be an A-definable set. Let M be a copy of the prime model over A.

Definition 6.1. [27] A type $p(\bar{x})$ over A is said to be strongly internal to a set X if for any $b, b' \models p(\bar{x})$, $b' \in \text{dcl}(A, X, b)$.

Example 6.2. Let Y be internal to X as in the previous section. Then the type of any fundamental system for the data (Y, X) is strongly internal to the set X in the sense of [27].

Definition 6.3. [27] A type $p(\bar{x})$ over A is said to be weakly orthogonal to X if for any tuple $\bar{c}$ from X, letting $r(\bar{x}) = tp(\bar{c}/A)$, $p(\bar{y}) \cup r(\bar{x})$ extends to a complete type over A in the variables $\bar{y}\bar{x}$. Equivalently, $p(\bar{x})$ has a unique extension to a type over (the not small set of parameters) $\text{dcl}(A, X)$.

Fact 6.4 ([27] Lemma 2.2 i)). *The type q is weakly orthogonal to X if and only if for any realization b of q, $\text{dcl}(A, b) \cap X(M) = X(\text{dcl}(A, b)) = X(A)$.*

Definition 6.5. [27] A definable Galois extension $A \subseteq B$ relative to X is a dcl-closed finitely dcl-generated extension $B = \text{dcl}(A, \bar{b})$ where $\bar{b}$ is a realization of a type $p(\bar{x})$ strongly internal and weakly orthogonal to X.

Fact 6.6. *Let $p(\bar{y})$ be a type over A which is strongly internal and weakly orthogonal to a set X defined over A.*

1. *The type $p(\bar{y})$ is atomic, and so is realized in M.*
2. *If X is strongly minimal, then $X(M) = X(\text{acl}(A))$.*
3. *If X is strongly minimal, then every realization of $p(\bar{y})$ in M is contained in the minimal closure of A (the intersection of all embeddings of M into itself.)*

Proof. We give a mixture of references and proofs.

1. This is shown in [27] and [19]. We will essentially give the argument in the next section.
2. This is Lemma 2.4 of [19]. By strong minimality, the only realizations of X which are atomic are algebraic over A hence $X(M) = X(\text{acl}(A))$.
3. This is in [23]. Let $\bar{b_1}$ be a realization of the type $p(\bar{y})$ in M. Let $\gamma : M \to M$ be an elementary embedding. So $\gamma(M) \prec M$. Let $\bar{b}_0$ be a realization of $p(\bar{y})$ in $\gamma(M)$. Then $\bar{b}_1 \in \text{dcl}(A, X, \bar{b}_0)$. It quickly follows that $\bar{b}_1 \in \text{dcl}(A, X(M), \bar{b}_0) = \text{dcl}(A, X(\text{acl}(A)), \bar{b}_0)$. As $\text{acl}(A)$ is contained in

the minimal closure of A, $\operatorname{dcl}(A, X(\operatorname{acl}(A)), \bar{b}_0)$ is contained in the image of γ. So b_1 is in the image of all such embeddings.

□

We finish this subsection with the connection between model theoretic primitives and definable Galois extensions which we will use in Remark 8.3:

Theorem 6.7. [26] *Let $\bar{M}$ be a monster model of a t.t. theory T eliminating imaginaries, and let A be a (small) set of parameters.*

(1) *Let Q be an A-definable right torsor for an A-definable group H, and let $b \in Q(\mathcal{U})$. Then, $B = \operatorname{dcl}(A, b)$ is a Galois extension of A relative to H in the sense of Definition 6.5 if and only if $H(B) = H(A)$.*

(2) *Let B/A be a definable Galois extension; namely, $B = \operatorname{dcl}(A, b)$ with $q = tp(b/A)$ strongly internal and weakly orthogonal to some A-definable set. Let $Q \circlearrowleft H$ be the associated A-definable right torsor given by the torsor theorem (i.e., part (2) of Fact 6.6). Suppose H is subgroup of an A-definable group G. Then, there is an A-definable right torsor W for G such that Q is A-definably isomorphic to an orbit of H on W and the Galois extension B/A can be generated by a (W, G, H)-primitive over A.*

Moreover, the image of the class associated to the torsor Q under the natural map

$$\mathrm{H}^1_{\mathrm{def}}(A, H) \xrightarrow{\iota^1} \mathrm{H}^1_{\mathrm{def}}(A, G)$$

is trivial if and only if Q is A-definably isomorphic to an A-definable left coset of H in G. In this case, the Galois extension B/A can be generated by a (G, H)-primitive. This happens in particular when $\mathrm{H}^1_{\mathrm{def}}(A, G) = 1$.

7. The binding groupoid

Let Y and X be A-definable with Y internal to X. Let Z be the associated A-definable set of fundamental systems and by Lemma 5.9 let $f : Z \times X_1 \to Z$ be such that for any $b, b' \in Z$, there exists a unique $c \in X_1$ with $f(b, c) = b'$.

Remark 7.1. Note that from any pair of elements of Z we can define uniquely an element of X_1. We have a definable function $h : Z^2 \to X_1$. Then $f(y_1, h(y_2, y_3)) = y_4$ is a ternary function. If this ternary function satisfies two identities

(1) $f(y_1, h(y_2, y_2) = y_1 = f(y_2, h(y_2, y_1))$
(2) $f(y_1, h(y_2, f(y_3, h(y_4, y_5)))) = f(f(y_1, h(y_2, y_3)), h(y_4, y_5))$

we can recover uniquely the structure of a group on X_1 and a PHS on Z. There seems to be no reason in general for these identities to be satisfied, i.e. for (Z, X_1) to have the structure of a PHS. At the end of this subsection we address the case where (Z, X_1) is a PHS.

Lemma 7.2. *Let $b \in Z$. Then $tp(b/\operatorname{dcl}(A, X))$ is isolated.*

Proof. Let $p(y) = tp(b/\operatorname{dcl}(A, X))$.

As $M \prec \bar{M}$, there is some $b_0 \in Z(M)$ (assuming Z is nonempty). Note that $X(M)$ is a normal subset of M so M is atomic over $\operatorname{dcl}(A, X(M))$. Hence $tp(b_0/\operatorname{dcl}(A, X(M)))$ is isolated by a formula $\psi(y, \bar{d}_0)$ where $\bar{d}_0$ is a tuple from $\operatorname{dcl}(A, X(M))$. Let the realizations of this type be $Q_{\bar{d}_0}$.

Let $c \in X_1$ with $f(b_0, c) = b$. Then

$$\exists y_0 \in Z \ \psi(y_0, \bar{d}_0) \wedge f(y_0, c) = y_1$$

defines a nonempty definable subset of Z. Note that if $c \in \operatorname{dcl}(A, X(M))$, then this formula is $\operatorname{dcl}(A, X(M))$-definable and is therefore realized in M.

It remains to show that this formula isolates $p(y)$. The formula is clearly over $\operatorname{dcl}(A, X)$ and b satisfies it, so it is in the type. Now let b' realize the above formula. Let b'_0 witness that

$$\psi(b'_0, d_0) \wedge f(b'_0, c) = b'.$$

Let $\sigma \in \operatorname{Aut}(\bar{M}/\operatorname{dcl}(A, X))$ take b_0 to b'_0. Then $\sigma(b) = f(\sigma(b_0), c) = f(b'_0, c) = b'$. It then follows that b' has the same type as b over $\operatorname{dcl}(A, X(M))$. So the type $p(y)$ is isolated over $\operatorname{dcl}(A, X(M))$ with tuple of parameters $\bar{d}_0, c$. □

Remark 7.3. First note that in the above proof $c \in \operatorname{dcl}(A, X(M))$ iff $b \in M$.

Secondly, by the normality of $\operatorname{dcl}(A, X(M))$, for any $\sigma \in \operatorname{Aut}(\bar{M}/A)$, then $\psi(y, \sigma(\bar{d}_0))$ isolates a complete type over $\operatorname{dcl}(A, X(\sigma(M)))$. Then the formula $\exists y_0 \in Z \ \psi(y_0, \sigma(\bar{d}_0)) \wedge f(y_0, \sigma(c)) = y_1$ isolates the type of some $b \in Z$ over $\operatorname{dcl}(A, X)$.

Definition 7.4. We define two A-definable binary relations on Z^2 as follows.

(1) Say (b_1, b'_1) is E_{int}-equivalent to (b_2, b'_2) if there is a $c \in X_1$ such that $f(b_1, c) = b_2$ and $f(b'_1, c) = b'_2$. The int stands for intrinsic.
(2) Say (b_1, b'_1) is E_{ext}-equivalent to (b_2, b'_2) if there is a $c \in X_1$ such that $f(b_1, c) = b'_1$ and $f(b_2, c) = b'_2$. The ext stands for extrinsic.

Proposition 7.5. *Let $b \in Z$. By the preceding lemma $tp(b/\operatorname{dcl}(A, X))$ is isolated. Let d be a tuple from $\operatorname{dcl}(A, X)$ isolating the type $tp(b/\operatorname{dcl}(A, X))$. Let Q_d be the definable set of realizations of the type.*

(1) *Then E_{int} and E_{ext} restricted to Q_d^2 are definable equivalence relations.*
(2) *$H_d^+ := (Q_d)^2/E_{int}$ is a $\operatorname{dcl}(A, d)$-definable group with a A-definable action on Q_d abstractly isomorphic to the action of $Aut(Q_d/\operatorname{dcl}(A, X))$ on Q_d via elementary permutation.*
(3) *A descent argument shows that each H_d^+ is in definable bijection with a group H^+ defined over A. Moreover H^+ acts on Z on the left abstractly isomorphically to the action of $Aut(Z/\operatorname{dcl}(A, X))$ on Z.*
(4) *Define $O := Z/H^+ \subseteq \operatorname{dcl}(A, X)$. Let $\pi : Z \to O$ be the A-definable quotient map. For any $b \in Z$, $tp(b/\operatorname{dcl}(A, X))$ is isolated over $\pi(b) = d \in \operatorname{dcl}(A, X)$ by the formula $\pi(y) = d$.*
(5) *For any Q_d, $Q_{d'}$, define $H_{dd'} \subseteq X_1$ to be $c \in X_1$ such that $f(b, c) \in Q_{\bar{d}'}$ for any $b \in Q_d$. This set is $\operatorname{dcl}(A, d, d')$-definable. Write H_{dd} as H_d.*
(6) *$H_d \cong Q_d^2/E_{ext}$. Moreover H_d is an $\operatorname{dcl}(A, d)$ definable group acting via f on the right on Q_d giving (H^+, Q_d, H_d) the structure of a $\operatorname{dcl}(A, d)$-definable biprincipal homogeneous space.*
(7) *If $d \in O(A)$, then there is a $b \in Q_{\bar{d}}(M)$ and $\operatorname{dcl}(A, b)$ generates an H_d-definable Galois extension of A in M. Otherwise in general an element $b \in Q_{\bar{d}}(M)$ generates an H_d-definable Galois extension of $\operatorname{dcl}(A, d)$ in M.*

Proof. More or less as in [4] appendix B.

(1) For reflexivity of E_{int} on $Q_{\bar{d}}^2$: Let $b_1, b_1' \in Q_{\bar{d}}$. There exists a $c \in X_1$ such that $f(\bar{b}_1, c) = \bar{b}_1$. We need to show that this same c satisfies $f(\bar{b}_1', c) = \bar{b}_1'$. Let $\sigma \in \text{Aut}(\bar{M}/\operatorname{dcl}(A, X))$ with $\sigma(\bar{b}_1) = \bar{b}_1'$. Apply σ to $f(\bar{b}_1, c) = \bar{b}_1$.

For symmetry of E_{int} on $Q_{\bar{d}}^2$ let (b_1, b_1') be E_{int}-related to (b_2, b_2') via $c \in X_1$. There exists $c' \in X_1$ such that $f(b_2, c') = b_1$. We want to show $f(b_2', c') = b_1'$. Again, let $\sigma \in \text{Aut}(\bar{M}/\operatorname{dcl}(A, X))$ with $\sigma(\bar{b}_1) = \bar{b}_1'$. As $f(b_1, c) = b_2$ and $f(b_1', c) = b_2'$, $\sigma(b_2) = \sigma(f(b_1, c)) = f(\sigma(b_1), c) = f(b_1', c) = b_2'$. Then applying σ to $f(b_2, c') = b_1$, we see $f(b_2', c') = \sigma(f(b_2, c')) = \sigma(b_1) = b_1'$ as desired. Transitivity follows similarly.

That E_{ext} is an equivalence relation on $Q_{\bar{d}}^2$ is immediate.

(2) The set $Q_{\bar{d}}$ is $\operatorname{dcl}(A, \bar{d})$ definable and E_{int} is A-definable so the quotient H_d^+ is $\operatorname{dcl}(A, \bar{d})$-definable by EI. Define a map $\iota_d : \text{Aut}(Q_d/\operatorname{dcl}(A, X)) \to H_d^+$ by sending $\sigma \mapsto [(\sigma(b), b)]$ for some $b \in Q_d$. This is well defined independent of the choice of b: If $b_1, b_2 \in Q_d$, there is $c \in X_1$ with $f(b_1, c) = b_2$, then also $f(\sigma(b_1), c) = \sigma(b_2)$. So $[(\sigma(b_1), b_1)] = [(\sigma(b_2), b_2)]$. It is surjective by strong homogeneity of the monster model over stably embedded sets.

It is injective: Let $\sigma_1, \sigma_2 \in \text{Aut}(Q_d/\operatorname{dcl}(A, X))$ with the associated classes $[(\sigma_1(b), b)] = [(\sigma_2(b), b)]$. This means that there exists a $c \in X_1$ such that $f(b, c) = b$ and $f(\sigma_1(b), c) = \sigma_2(c)$. Then $b = f(b, c) = \sigma_1^{-1}\sigma_2(b)$. So $\sigma_1(b) = \sigma_2(b)$. Since an elementary permutation σ of $Q_{\bar{d}}$ fixing $\operatorname{dcl}(A, X)$ is uniquely determined by it's action on any $b \in Q_d$, we have $\sigma_1 = \sigma_2$.

Let $\pi_{int} : Q_d^2 \to H_d^+$ be the definable quotient map given by EI. The bijection $\iota_d : \text{Aut}(Q_d/\operatorname{dcl}(A, X)) \to H_d^+$ induces a group structure on H_d^+. It is an exercise to check that the group structure is defined by the following formula in the variables x_1, x_2, x_3 ranging over H_d^+:

$\exists y_1, y_2, y_3 \in Q_d\ \pi_{int}(y_2, y_1) = x_2 \wedge \pi_{int}(y_3, y_2) = x_1 \wedge \pi_{int}(y_3, y_1) = x_3$.

The action of H_d^+ on Q_d is A-definable and agrees with the action of $\text{Aut}(Q_{\bar{d}}/\operatorname{dcl}(A, X))$: Let $[(b', b)] \in H_{\bar{d}}^+$ and $b'' \in Q_d$. Then we evaluate $[(b', b)]b''$ as the unique b''' such that $\pi(b''', b'') = [(b', b)]$. This is A-definable. Let x and y, y_1 be variables ranging over $H_{\bar{d}}^+$ and Q_d respectively, then the action $xy = y_1$ is A-definable by the formula $\pi_{int}(y_1, y) = x$. It is clear that this agrees with the action of $\text{Aut}(Q_{\bar{d}}/\operatorname{dcl}(A, X))$ on the left.

(3) We adapt the argument showing G is F-definable from [4] Theorem B.1. Let $H_{d_1}^+$ and $H_{d_2}^+$ be two such groups. Note that $\text{Aut}(Q_{d_1}/\operatorname{dcl}(A, X))$ and also $\text{Aut}(Q_{d_2}/\operatorname{dcl}(A, X))$ are both canonically isomorphic to $\text{Aut}(Z/\operatorname{dcl}(A, X))$ as any such elementary permutation of Z, Q_{d_1}, or Q_{d_2} is entirely determined by the image of any single fundamental system b. So making the necessary identifications, we have a map $\iota_{d_2} \circ \iota_{d_1}^{-1} : H_{d_1}^+ \to H_{d_2}^+$. It is easy to verify that this map sends $[(b_1, b_1')] \in H_{d_1}^+$ to $[(b_2, b_2')] \in H_{d_2}^+$ such that $\exists c \in X_1$ with $f(b_1, c) = b_2$ and $f(b_1', c) = b_2'$. Then $\iota_{d_2} \circ \iota_{d_1}^{-1}(x_1) = x_2$ is definable over $d_1 d_2$ via $\exists y_1, y_2 \in Q_{d_1}, \exists y_3, y_4 \in Q_{d_2},\ \exists c \in X_1,\ \pi_{\text{int}}(y_2, y_1) = x_1 \wedge \pi_{\text{int}}(y_4, y_3) = x_2 \wedge f(y_2, c) = y_4 \wedge f(y_1, c) = y_3$.

Let $\phi(x)$ isolate $tp(d_0/A)$ as in the proof of Lemma 7.2 and D be the set of realizations. Note that each d is a concatenation of d_0 and an element $c \in X_1$. Consider the A-definable set $x_0 \in D \wedge z \in X_1 \wedge x \in H_{x_0 z}^+$ i.e. the parameterized family of H_d^+. Define a relation via: $(x_0, z, x) \equiv (x_0', z', x')$ if $\iota_{x_0' z'} \circ \iota_{x_0 z}^{-1}(x) = x'$. It is clear that this identifies distinct elements of

the different H_d^+ which correspond to the same elementary permutation of $\mathrm{Aut}(Z/\operatorname{dcl}(A,X))$ from which we see that it is an A-definable equivalence relation. In fact it is the equivalence relation whose classes are fibers of the piecewise map $\sqcup_d \iota_d^{-1} : \sqcup H_d^+ \to \mathrm{Aut}(Z/\operatorname{dcl}(A,X))$. Let H^+ be the quotient. It is clear that H^+ acts on Z on the right A-definably isomorphically to the action of $\mathrm{Aut}(Z/\operatorname{dcl}(A,X))$ on Z.

(4) That O is contained in $\operatorname{dcl}(A,X)$ is because the action of H^+ is the action of $\mathrm{Aut}(Z/\operatorname{dcl}(A,X))$ on Z. The orbits (equivalently, the fibers of $\pi : Z \to O$) are exactly the sets Q_d.

(5) This is a definition.

(6) Define $h_{\mathrm{ext}} : Q_d^2 \to H_d$ via $(b,b') \mapsto c$ such that $f(b,c) = b'$. For $b \in Q_d$, define $\rho_b : \mathrm{Aut}(Q_d/\operatorname{dcl}(A,X)) \to H_d$ via $\sigma \mapsto c_\sigma \in H_d$ such that $f(b,c_\sigma) = \sigma(b)$. It is clear that ρ_b is a bijection and that the identity automorphism gets sent to an element in $\operatorname{dcl}(A,d)$. The formula $\exists y \in Q_d f(f(y,x_1),x_2) = f(y,x_3)$ defines the induced group operation on H_d and the action of H_d on the right on Q_d is given by f. Let $\sigma \in \mathrm{Aut}(Q_d/\operatorname{dcl}(A,X))$, $c \in H_d$, and $b \in Q_d$. The equality $\sigma(f(b,c)) = f(\sigma(f),c)$ shows the compatibility of the biprincipal homogeneous space as the action of H^+ is via elementary permutation.

(7) By Theorem 6.7 part 1, as (Q_d, H_d) is a $\operatorname{dcl}(A,d)$ definable torsor, for $b \in Q_d$, $\operatorname{dcl}(A,b)$ is a definable Galois extension of $\operatorname{dcl}(A,d)$ if $H_d(\operatorname{dcl}(A,b)) = H_d(\operatorname{dcl}(A,d))$. But this follows: Let $g(b) = c \in H_d(\operatorname{dcl}(A,b))$. Let $\sigma \in \mathrm{Aut}(\bar{M}/\operatorname{dcl}(A,d))$, then σ fixes $tp(b/\operatorname{dcl}(A,X))$ which we called Q_d. As $g(y) = \sigma(c) \in tp(b/\operatorname{dcl}(A,X))$ so $\sigma(c) = c$, so $c \in \operatorname{dcl}(A,d)$.

Note that if $d \in O(A)$ then $\exists b \in Q_d(M)$ (See Remark 7.3) and then $\operatorname{dcl}(A,b)$ is a definable Galois extension of A.

□

8. Multiplicity of definable Galois extensions for internality data

In this section we prove an analogue of the following result in the totally transcendental setting.

Theorem 8.1. [3] *Let Y be an ordinary homogeneous linear differential equation over a differential field K. Assume that there is at least one nontrivial Picard-Vessiot extension L/K for K in K^{diff}, a differential closure of K. Let $H(C_{K^{\mathrm{diff}}})$, a linear algebraic group defined over C_K, be the (extrinsic) Picard-Vessiot differential Galois group of the extension L/K. Then the usual algebraic Galois cohomology*

$$\mathrm{H}^1_{\mathrm{alg}}(C_{K^{\mathrm{diff}}}/C_K, H(C_K))$$

classifies (counts) exactly the set of distinct Picard-Vessiot extensions for Y in K^{diff} with the base point of the pointed cohomology set corresponding to L/K.

Let (Y,X) be fixed internality data. Construct X_1 and Z as in the construction of the binding groupoid. We now focus on the case where X_1 is a A-definable group and Z is a A-definable right PHS for X_1.

Theorem 8.2. *Let Z be a right A-definable PHS for an A-definable group X_1.*

(1) *E_{int} and E_{ext} are then A-definable equivalence relations on Z^2.*

(2) *Z^2/E_{int} is an A-definable group which acts on the left on Z as the set of all permutations commuting with the right action of X_1. It has H^+ as the A-definable subgroup of elementary permutations commuting with the right action of X_1.*

(3) *Assume there exists at least one $b_0 \in Z(M)$ with $\pi(b_0) = d_0 \in O(A)$ and extrinsic group H_{d_0}. Then the set of definable Galois extensions in M generated by elements of $Z(M)$ embeds in $\mathrm{H}^1_{\mathrm{def}}(M/A, H_{d_0}(M))$. If the cohomology $\mathrm{H}^1_{\mathrm{def}}(M/A, X_1(M))$ is trivial then this embedding is an isomorphism. This extends Proposition 3.2.19 of* [12].

Proof. (1) Reflexivity of E_{int}: Let $e \in X_1$ be the identity. Then $f(b_1, e) = b_1$ and $f(b_1', e) = b_1'$ for any $(b_1, b_1') \in Z^2$. Symmetry of E_{int}: Let (b_1, b_1') be E_{int}-equivalent to (b_2, b_2') witnessed by $c \in X_1$. Then $c^{-1} \in X_1$ witnesses that (b_2, b_2') is E_{int}-equivalent to (b_1, b_1'). Transitivity of E_{int}: Let (b_1, b_1') be E_{int}-equivalent to (b_2, b_2') witnessed by $c_{12} \in X_1$. Let (b_2, b_2') be E_{int}-equivalent to (b_3, b_3') witnessed by $c_{23} \in X_1$. Then the product in X_1 of $c_{12}c_{23}$ witnesses that (b_1, b_1') is E_{int}-equivalent to (b_3, b_3').

Reflexivity, symmetry, and transitivity of E_{ext} are all immediate using the fact that Z is a PHS for X_1.

(2) The definition of E_{int} and an easy computation show that Z^2/E_{int} determines uniquely a permutation of Z commuting with the right action of X_1. It is clear that H^+ is just those permutations preserving type over $\mathrm{dcl}(A, X)$ and is defined by exactly the formula in the variable $x \in Z^2/E_{int}$ given as $\exists y_1, y_2 \in Z\ \exists d \in D \times X_1\ \pi_{int}(y_1, y_2) = x \wedge y_1 \equiv_d y_2$ where $y_1 \equiv_d y_2$ says y_1 and y_2 both satisfy same type over d, i.e. the both satisfy the formula given in Remark 7.3. Recall that D is defined in the proof of part 3) of Proposition 7.5.

The induced group operation on Z^2/E_{int} is A-definable: The product of two E_{int}-classes $[(b_1, b_1')]$ and $[(b_2, b_2')]$ is computed as follows. As Z is a PHS for X_1 there is a unique $c \in X_1$ such that $f(b_2, c) = b_1'$. Then $[(b_2, b_2')] = [(b_1', f(b_2', c))]$. So $[(b_1, b_1')][(b_2, b_2')] = [(b_1, b_1')][(b_1', f(b_2', c))]$ we then define $[(b_1, b_1')][(b_1', f(b_2', c))] := [(b_1, f(b_2', c))]$.

The action of Z^2/E_{int} or H^+ on Z is defined as before via $[(b, b')]b' := b$.

(3) We adapt Proposition 3.2.19 of [12].

Let $b_1 \in Z(M)$, such that $\mathrm{dcl}(A, b_1)$ is a definable Galois extension for the equation. By the last part of Proposition 7.5, $f(b_1) = d_1 \in O(A)$ and therefore $H_{d_1,d_0}(M)$ is an A-definable right PHS for $H_{d_0}(M)$ which gives the desired class. Recall that $H_{d_1,d_0}(M)$ is defined as

$$\{c \in X_1(M)\ :\ \exists b_1 \in Q_{d_1}(M),\ \exists b_0 \in Q_{d_0}(M),\ f(b_1, c) = b_0\}$$

It is clear that if $b_1, b_2 \in Z(M)$ generate the same definable Galois extension then there is a $c \in X_1(A)$ with $f(b_1, c) = b_2$. This yields an A-definable bijection between the right PHSs $H_{d_1,d_0}(M)$ and $H_{d_2,d_0}(M)$, and so the resulting cohomology classes are isomorphic, giving injectivity.

Note also that $H_{d_0,d_1}(M)$ is an A-definable left PHS for $H_{d_0}(M)$ so we could phrase everything in terms of left PHSs/cohomology if we wanted to.

For the moreover clause we need the long exact sequence in cohomology. Let $P(M)$ be a right A-definable PHS for $H_{d_0}(M)$. By the assumption that $\mathrm{H}^1_{\mathrm{alg}}(M/A, X_1(M))$ is trivial, the class associated to $P(M)$ must be in the

image of the connecting map δ^1:

$$1 \to H_{d_0}(A) \to X_1(A) \to (X_1/H_{d_0})(A) \xrightarrow{\delta^1}$$
$$\mathrm{H}^1_{\mathrm{alg}}(M/A, H_{d_0}(M)) \to 1 \ (= \mathrm{H}^1_{\mathrm{alg}}(M/A, X_1(M)))$$

To see this, note first that the action of H_{d_0} gives an A-definable equivalence relation on the A-definable group X_1. So by EI, the quotient is an A-definable set of left cosets (X_1/H_{d_0}). Note also the quotient map π is a A-definable function. Given an A-point, $p \in (X_1/H_{d_0})(A)$, we have that $\mathrm{dcl}(A, p) = A$. The preimage $\pi^{-1}(p)$ is then an A-definable left coset of $H_{d_0}(M)$ in $X_1(M)$ which is then a right A-definable PHS for $H_{d_0}(M)$.

So the assumption requiring the vanishing of the cohomology together with the long exact sequence implies that any definable right PHS for $H_d(M)$ must be a A-definable coset of $H_d(M)$ in $X_1(M)$. But this is exactly something of the form $H_{d_1,d_0}(M)$ because $Z(M)$ is a right PHS for $X_1(M)$: Take an element c in this coset. Apply the inverse c^{-1} to b_0 to get a new solution b_1 with $f(b_1) = d_1$. Then the coset is actually equal to $H_{d_1 d_0}(M)$ as they intersect nontrivially. In the previous step we use that there is already one proper definable Galois extension, so that we may take such an element b_0.

□

Remark 8.3. In general there seems to be no reason why X_1 needs to carry a group structure making (Z, X_1) into an A-definable PHS with action f. However this assumption is satisfied whenever the internality data we begin with (Y, X) has the form of an A-definable PHS.

Very often the internality data given is in the form of an equation on a group. By this, concretely we mean an equation coming from a G-primitive in the model-theoretic sense of Pillay-Sokolovic or V-primitive in the model-theoretic sense defined in [12] as in Definition 4.5 above. Indeed, starting with a fixed extension, Theorem 6.7 [26] tells us exactly that we can find a torsor which gives rise to the extension i.e. the extension has a generator which is a primitive element in this sense. So in practice, this assumption is very often satisfied. In the differential algebraic setting of $T = \mathrm{DCF}_{0,m}$, such equations are the logarithmic differential equations of Kolchin on a (differential) algebraic groups [26].

9. Colimit formulas and boundedness for differential fields

We begin by reviewing the basic relationships between algebraic algebraic Galois cohomology and Galois extensions. We will then survey, prove, and propose some analogous statements in the differential algebraic setting.

Fact 9.1. [25] *Let F be a perfect field. The following are equivalent:*

(1) $F = F^{\mathrm{alg}}$.
(2) *For any finite algebraic group G defined over F,*
$$\mathrm{H}^1_{\mathrm{alg}}(F^{\mathrm{alg}}/F, G(F^{\mathrm{alg}})) = 1.$$
(3) *For any algebraic group G defined over F,*
$$\mathrm{H}^1_{\mathrm{alg}}(F^{\mathrm{alg}}/F, G(F^{\mathrm{alg}})) = 1.$$

We next recall the usual colimit formula for Galois cohomology.

Fact 9.2. [25] *Let F be a perfect field and G be an algebraic group defined over F. Then*

$$\mathrm{H}^1_{\mathrm{alg}}(F^{\mathrm{alg}}/F, G(F^{\mathrm{alg}})) = \varinjlim \mathrm{H}^1_{\mathrm{alg}}(E/F, G(E))$$

where the limit is taken over the directed system of finite Galois extensions E of F.

We finally recall the notion of boundedness of a field and some cohomological consequences.

Fact 9.3. [25] *A perfect field F is said to be bounded if any of the following equivalent conditions hold*

1. *F has finitely many Galois extensions of each finite degree.*
2. *For any linear algebraic group G over F, $\mathrm{H}^1_{\mathrm{alg}}(F^{\mathrm{alg}}/F, G(F^{\mathrm{alg}}))$ is finite.*
3. *For any linear algebraic group G over F, there is a finite Galois extension E of F such that $\mathrm{H}^1_{\mathrm{alg}}(E/F, G(E)) \cong \mathrm{H}^1_{\mathrm{alg}}(F^{\mathrm{alg}}/F, G(F^{\mathrm{alg}}))$.*

We now put together some results in the literature which give differential algebraic analogues of the above, and propose notions of boundedness suitable for a differential field.

In the rest of this section K is a differential field of characteristic 0 with algebraically closed field of constants, $C_K = (C_K)^{\mathrm{alg}}$. We use the conventions of [15] for finite and infinite Picard-Vessiot (PV) extensions. Namely a PV extension L of K is an extension generated by fundamental systems (in K^{diff}) of solutions to ordinary homogeneous linear differential equations over K. If L is generated over K as a differential field by the fundamental system of a single equation (it is equivalent to ask for finitely many) we will call L a PV extension of finite type.

We recall the definition of the sequence of Picard-Vessiot envelopes of a differential field K and the Picard-Vessiot closure from [15] for example. Let K^{PV_1} be the union of all Picard-Vessiot extensions of K. (It is enough to ask for them inside of K^{diff}.) As is well known, see Example 3.33 of [11], K^{PV_1} may itself not be closed in K^{diff}, and the union of all PV extensions of K^{PV_1} in K^{diff} we write as K^{PV_2} and so forth. We define $K^{PV_\infty} = \cup_n K^{PV_n}$ which is itself PV closed and is called the Picard Vessiot closure of K. As in [15], this yields a chain of normal differential subfields of the minimal closure of K,

$$K \subseteq K^{PV_1} \subseteq K^{PV_2} \subseteq ...K^{PV_\infty} \subseteq \mathrm{mcl}(K) \subseteq K^{\mathrm{diff}}.$$

There are then at least three intermediate differential fields between K and K^{diff} (or intermediate sets between A and M) which each, in different senses, generalize the algebraic closure. Namely, $K^{PV_\infty} \subseteq \mathrm{mcl}(K) \subseteq K^{\mathrm{diff}}$.

The following differential algebraic version of Fact 9.1 was shown in both [21] and [16]: In fact, [21] covers the generalized strongly normal theory, while [16] works in multiple commuting derivations in the linear case with an additional assumption.

Theorem 9.4. [21] *Let K be a differential field. The following are equivalent:*

1. *K is algebraically closed and PV-closed.*
2. *For any G a linear algebraic group in the constants then*

$$H^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}})) = 1$$

3. *For any G be a linear differential algebraic group over K we have*

$$H^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}})) = 1$$

Remark 9.5. If $K = K^{\mathrm{diff}}$, then from the definition, $H^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}})) = 1$ for any linear differential algebraic group G over K. We show below in Proposition 9.9 (in the more general model theoretic context) that if $K = \mathrm{mcl}(K)$ then again $H^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}})) = 1$ for any linear differential algebraic group G over K. However, the above Theorem 9.4 together with abundant examples where the inclusions $K^{PV_\infty} \subseteq \mathrm{mcl}(K) \subseteq K^{\mathrm{diff}}$ are strict show that the converses are false.

Corollary 9.6. [14] *Let K be a differential field and G a linear algebraic group in the constants of K. Then*

$$H^1_\delta(K^{PV_\infty}/K, G(K^{PV_\infty})) \cong H^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}})).$$

Proof. Immediate from Theorem 4.2 applied to $K \subseteq K^{PV_\infty} \subseteq K^{\mathrm{diff}}$ and Theorem 9.4. □

In preparation to show some differential algebraic versions of the colimit formula of Fact 9.2, we make some remarks about various kinds of subfields of K^{PV_∞}. We will comment on finitely differentially generated subfields, normal subfields, Picard-Vessiot extensions, and iterated Picard-Vessiot extensions.

We contine to follow [15] for notation and cite results from [10].

An iterated PV (IPV) extension of K is a differential field extension $K \subseteq L \subseteq K^{PV_\infty}$, such that there exists an infinite sequence of proper Picard-Vessiot extensions L_i/L_{i-1}, not necessarily of finite type, such that L is the union of that chain, or there exists a finite such chain $K = L_0 \subseteq L_1 \subseteq L_2 \subseteq \cdots L_n$ with $L_n = L$.

Theorem 2 of [10] says that $K \subseteq L \subseteq K^{PV_\infty}$ is finitely differentially generated if and only if L is a differential subfield of an IPV extension which is the union of a finite chain where each L_i/L_{i-1} is a PV extension of finite type. We will call IVP extensions of this form IVP extensions of finite type.

Proposition 3.2 of [15], shows that any normal extension, $K \subseteq L \subseteq K^{PV_\infty}$, is an IPV extension.

Lemma 2 of [10] says that the compositum of two (or finitely many) IPV extensions (in our sense) is again an IPV extension. The proof shows that if the two extensions are of finite type the compositum will be of finite type. So normal IPV extensions form a directed system whose union is K^{PV_∞}. This statement is tautological because K^{PV_∞} is itself a normal IPV extension. We address this as follows:

Firstly, restricting to finitely differentially generated normal IPV extensions yields exactly the family of PV extensions of finite type:

Lemma 9.7. *Let $K \subseteq L \subseteq K^{PV_\infty}$. The following are equivalent:*

1. *L is normal and finitely differentially generated over K.*
2. *L is a normal IPV extension of finite type over K.*
3. *L is a PV extension of finite type over K.*

Proof. We show (1) implies (2) implies (3). The remaining (3) implies (1) is immediate.

Let L be normal and finitely differentially generated over K. As we assumed $C_K = C_K^{\mathrm{alg}}$, L is then a strongly normal extension of K.

As L is finitely differentially generated, Theorem 2 of [10] says that L is a differential subfield of an IPV extension of finite type. In the proof of Theorem 2 of [10], an additional assumption of normality yields that L is exactly an IPV extension of

K of finite type. But then the Galois group of L must have a composition series of linear algebraic groups and so must be linear.

By Fact 1.20 of [14], a strongly normal extension with a linear algebraic Galois group must be a PV extension of finite type.

□

Proposition 9.8. *Let G be a linear differential algebraic group defined over K. Then*

$$\varinjlim H^1_\delta(L/K, G(L)) \cong H^1_\delta(K^{PV_1}/K, G(K^{PV_1}))$$

where the limit is again taken over the family of normal, finitely differentially generated extensions $K \subseteq L \subseteq K^{PV_\infty}$ or equivalently, the family of Picard-Vessiot extensions of finite type.

Proof. The normal, finitely differentially generated extensions $K \subseteq L \subseteq K^{PV_\infty}$ form a directed system under inclusion. By the previous Lemma 9.7, these are exactly the PV-extensions of finite type, so their direct limit is exactly K^{PV_1}. We then apply Lemma 4.3. □

Lemma 9.7 above, verifies the remark (immediately following Definition 4) of [10], that an IPV extension of finite type which is not a PV extension of finite type cannot be normal. Therefore, to obtain a colimit formula for K^{PV_∞} this we must work with the set of normal closures of IPV extensions of K of finite type:

Proposition 9.9. *Let G be a linear differential algebraic group G over K. Then*

$$\varinjlim H^1_\delta(L/K, G(L)) \cong H^1_\delta(K^{PV_\infty}/K, G(K^{PV_\infty}))$$

where the limit is taken over the family of normal closures of IPV extensions of finite type.

Proof. By Proposition 3.2 of [15], the normal closure of an IPV extension of finite type is again an IPV extension (although not necessarily of finite type). Then by Lemma 2 of [10], the normal closures of IPV extensions of finite type form a directed system whose direct limit is K^{PV_∞}. We then apply Lemma 4.3. □

From Corollary 9.6 and from Proposition 9.9 we obtain the following analogue of Fact 9.2, the usual colimit formula in the differential algebraic setting:

Proposition 9.10. *Let G be a linear differential algebraic group defined over a differential field K. Then*

$$\varinjlim \mathrm{H}^1_\delta(L/K, G(L)) \cong \mathrm{H}^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}}))$$

where the limit is taken over the directed system of normal closures of iterated PV extensions of K of finite type.

We then propose the following definition of boundedness for a differential field which can be added to the list of such notions in Section 2.3 of [14]:

Definition 9.11. A differential field K is said to be bounded with respect to a differential algebraic group G defined over K if there is an iterated PV extension of finite type, whose normal closure L, satisfies

$$\mathrm{H}^1_\delta(L/K, G(L)) \cong \mathrm{H}^1_\delta(K^{\mathrm{diff}}/K, G(K^{\mathrm{diff}})).$$

Remark 9.12. Note that Magid's example in Section 4 of [10] of an normal IPV extension E, is remarked to be exactly the normal closure of an IPV extension of finite type, E_0.

10. The minimal and definable Galois closures and colimit formulas for definable Galois cohomology

Motivated by the above observations in the differential algebraic setting, we give some analogous results in the model theoretic setting. Let T be a t.t. theory with EI. Let A be a small set of parameters and M a prime model over A. Let X be an A-definable set.

We make the following definitions and adjustments to the terminology to be consistent with the previous section.

Definition 10.1. Let A be a dcl-closed set of parameters. Let X be an A-definable set.

(1) In this section, a definable Galois extension over A relative to X as defined in Definition 6.5 will be called a definable Galois extension of finite type.
(2) By a definable Galois extension $A \subseteq B \subseteq M$ with respect to X we mean a union of definable Galois extensions of finite type over A relative to X.
(3) By A^{XDG_1} we mean the union of all X-definable Galois extensions $A \subseteq B$ in M.
(4) Having defined A^{XDG_n} we define $A^{\mathrm{XDG}_{n+1}}$ to be the union of all X-definable Galois extensions $A^{\mathrm{XDG}_n} \subseteq B$ in M.
(5) By A^{XDG_∞} we mean $\cup_n A^{\mathrm{XDG}_n}$.
(6) An iterated definable Galois extension $A \subseteq B \subseteq M$ relative to X is a dcl-closed subset such that either there exists an infinite sequence of dcl-closed subsets $A = B_0 \subseteq B_1 \subseteq \cdots$ with each B_i a proper definable Galois extension of B_{i-1} such that $B = \cup_i B_i$, or there exists a finite such sequence $A = B_0 \subseteq B_1 \subseteq \cdots B_n = B$. If B is the union of a finite such chain where each B_i/B_{i-1} is a definable Galois extension of finite type, we call B an iterated definable Galois extension of finite type of A relative to X.

Lemma 10.2. *Let G be an A-definable group. Let P be an A-definable PHS for G. If $G(M) = G(\mathrm{mcl}(A))$ then $P(M) = P(\mathrm{mcl}(A))$.*

Proof. By definition P is nonempty and therefore has a point p in M. Let $\gamma : M \to M$ be any A-elementary embedding. $\gamma(M) \prec M$ must also have a realization of P, p_1. By assumption $G(M) = G(\mathrm{mcl}(A)) = G(\gamma(M))$. Both p and p_1 are realizations of P in M and so there is a $g \in G(\mathrm{mcl}(A))$ with $p_1 g = p$. Hence $p \in \mathrm{dcl}(p_1, g) \subseteq \gamma(M)$. Since γ was arbitrary, $p \in \mathrm{mcl}(A)$. □

Remark 10.3. If G is strongly minimal or lives on a strongly minimal set then $G(M) = G(\mathrm{mcl}(A))$. For example, let K be a differential field of characteristic 0 and the ambient theory be the theory of differentially closed fields of characteristic 0, then one can take G to be an algebraic group defined over the constants C_K and the above condition will be satisfied.

Lemma 10.2 has the following consequences:

Theorem 10.4. *Let G be an A-definable group with $G(M) = G(\mathrm{mcl}(A))$. Then*

(1) *The definable Galois cohomology of G over $\mathrm{mcl}(A)$ is trivial:*

$$\mathrm{H}^1_{\mathrm{def}}(M/\mathrm{mcl}(A), G(M)) = 1.$$

(2) *Using the short exact sequence*

$$\mathrm{H}^1_{\mathrm{def}}(\mathrm{mcl}(A)/A, G(\mathrm{mcl}(A))) \cong \mathrm{H}^1_{\mathrm{def}}(A, G)$$

Namely, all of the definable Galois cohomological information for G in M is contained in $\mathrm{mcl}(A)$.

(3) *Any definable Galois extension relative to G is contained in* $\mathrm{mcl}(A)$.

Proof. (1) Let P be an A-definable right PHS for G. By Lemma 10.2, P has an $mcl(A)$ point so by definition $\mathrm{H}^1_{\mathrm{def}}(M/\mathrm{mcl}(A), G(M)) = 1$.

(2) By Theorem 4.2, the intermediate extension $A \subseteq \mathrm{mcl}(A) \subseteq M$ induces a short exact sequence in cohomology

$$1 \to \mathrm{H}^1_{\mathrm{def}}(\mathrm{mcl}(A)/A, G(\mathrm{mcl}(A))) \to \mathrm{H}^1_{\mathrm{def}}(A, G) \to \mathrm{H}^1_{\mathrm{def}}(M/\mathrm{mcl}(A), G(M)).$$

As the last term of the above sequence is 1 we have

$$\mathrm{H}^1_{\mathrm{def}}(\mathrm{mcl}(A)/A, G(\mathrm{mcl}(A))) \cong \mathrm{H}^1_{\mathrm{def}}(A, G).$$

(3) Let $A \subseteq B = \mathrm{dcl}(A, b)$ be a definable Galois extension relative to G. Let (Q, H) be the associated A-definable torsor. As H lives on G, $H(M) = H(\mathrm{mcl}(A))$. Then by Lemma 10.2, $Q(M) = Q(\mathrm{mcl}(A))$. As B is generated by a M-point of Q, B is contained in $\mathrm{mcl}(A)$. This is a slight strengthening of Fact 3.2 (3).

□

The next proposition generalizes the relationships between the series Picard-Vessiot envelopes, the Picard-Vessiot closure, the minimal closure, and differential closure of a differential field $K \subseteq K^{PV_1} \subseteq K^{PV_2} \cdots \subseteq K^{PV_\infty}$ as in [15].

Proposition 10.5. *Let A be a* dcl*-closed set of parameters. Let X be an A-definable set such that $X(M) = X(\mathrm{mcl}(A))$. Then each A^{XDG_n} is contained in* $\mathrm{mcl}(A)$ *and we have*

$$A \subseteq A^{XDG_1} \subseteq A^{XDG_2} \cdots \subseteq A^{XDG_\infty} \subseteq \mathrm{mcl}(A) \subseteq M.$$

Furthermore each A^{XDG_n} is normal.

Proof. This follows from Theorem 10.4 (3). Normality follows because each conjugate under $\mathrm{Aut}(M/A)$ of an X-definable Galois extension is again an X-definable Galois extension. □

We now strengthen the assumption $X(M) = X(\mathrm{mcl}(A))$ to the assumption that $X(M) = X(A)$. This is a generalization of the assumption that the field of constants of the base field be algebraically closed in the context of Picard-Vessiot differential Galois theory. It has the consequence that each X-definable Galois extension is then normal. An immediate consequence is that the family of X-definable Galois extensions and the family of normal infinite X-iterated definable Galois extensions form directed systems:

Lemma 10.6. *Let B and C be iterated definable Galois extensions of A of finite type relative to X in M. Then* $\mathrm{dcl}(B, C)$ *is again an iterated definable Galois extension of A of finite type relative to X in M.*

Proof. We follow Lemma 2 of [10]. Let $A = B_0 \subseteq B_1 \subseteq \cdots B_m = B$ and $A = C_0 \subseteq C_1 \subseteq \cdots C_n = C$ witness that B and C are iterated definable Galois extensions of finite type. The claim is proved by induction on n. If n is 0 then $C = A$ and $\mathrm{dcl}(B, C) = B$. Assume true for $n - 1$. So $\mathrm{dcl}(B_m, C_{n-1})$ is an iterated definable Galois extension of A. By assumption $C_n = \mathrm{dcl}(C_{n-1}, c_n)$ where $tp(c_n/C_{n-1})$ is weakly orthogonal to X and strongly internal to X. Then

by Fact 6.4, $tp(c_n/B_m, C_{n-1})$ is still weakly orthogonal to X as $X(M) = X(A)$. Adding parameters preserves strong internality to X. So $\mathrm{dcl}(B, C)$ is a definable Galois extension of $tp(c_n/B_m, C_{n-1})$ of finite type. So $\mathrm{dcl}(B, C)$ is still an iterated definable Galois extension of finite type. □

Lemma 10.7. *Let $X(M) = X(A)$. A normal iterated definable Galois extension B of A of finite type relative to X must be a definable Galois extension of finite type*

Proof. Let b dcl-generate B over A. Then by normality and strong homogeneity any other realization of the type of b over A must be contained in $\mathrm{dcl}(A, b, X(A))$. Again as $X(M) = X(A)$, $tp(b/A)$ is weakly orthogonal to X by Fact 6.4. □

Proposition 10.8. *Let G be an A definable group living on an A-definable set X. Assume $X(M) = X(A)$. Then*

$$\varinjlim \mathrm{H}^1_{\mathrm{def}}(B/A, G(B)) \cong \mathrm{H}^1_{\mathrm{def}}(A^{XDG_1}/A, G(A^{XDG_1}))$$

where the limit is taken over the family of normal iterated definable Galois extensions of finite type B of A relative to X, equivalently, the family of definable Galois extensions of finite type.

Proof. By Lemma 10.7, a normal iterated definable Galois extension of finite type B of A relative to X is a definable Galois extension of finite type. Then by Lemma 10.6, the family of definable Galois extensions of finite type B of A relative to X form a directed system. It is clear that their union is A^{XDG_1}. The result then follows from Lemma 4.3. □

As in the PV setting of the previous section, to obtain a colimit representation of $\mathrm{H}^1_{\mathrm{def}}(A^{XDG_\infty}/A, G(A^{XDG_\infty}))$ we must then work with the family of normal closures of iterated definable Galois extensions of finite type B of A relative to X.

Theorem 10.9. *Let G be an A definable group living on an A-definable set X. Assume $X(M) = X(A)$. Then*

$$\varinjlim \mathrm{H}^1_{\mathrm{def}}(B/A, G(B)) \cong \mathrm{H}^1_{\mathrm{def}}(A^{XDG_\infty}/A, G(A^{XDG_\infty}))$$

where the limit is taken over the family of normal closures of iterated definable Galois extensions of finite type B of A relative to X.

Proof. By Lemma 10.6, the family of normal closures of iterated definable Galois extensions of finite type B of A relative to X form a directed system. It is clear that their union is A^{XDG_∞}. The result then follows from Lemma 4.3. □

We then propose the following definition of a bounded set of parameters A.

Definition 10.10. Let X be an A-definable set satisfying $X(A) = X(M)$. Let G be an A-definable group living on X. The set of parameters A is said to be bounded with respect to G and X if there exists an iterated definable Galois extension of finite type relative to X whose normal closure B satisfies

$$\mathrm{H}^1_{\mathrm{def}}(B/A, G(B)) \cong \mathrm{H}^1_{\mathrm{def}}(A^{XDG_\infty}/A, G(A^{XDG_\infty})).$$

This definition is a generalization of Definition 9.11. However by Propositions 10.4 and 10.8, a general colimit formula and a definition of boundedness for all of $\mathrm{H}^1_{\mathrm{def}}(M/A, G(M))$, i.e., with reference to the extension M/A, would require and understanding of the difference between $\mathrm{H}^1_{\mathrm{def}}(A^{XDG_\infty}/A, G(A^{XDG_\infty}))$ and

$\mathrm{H}^1_{\mathrm{def}}(\mathrm{mcl}(A)/A, G(\mathrm{mcl}(A)))$. This is sensitive to the theory chosen and it is false in general that they are isomorphic:

In DCF_0 and the linear case of $\mathrm{DCF}_{0,m}$, the results of [21], [16], and [2] give the isomorphism in various cases. We note that difference algebra falls outside of the model theoretic totally transcendental setting of this paper. However, results of [2] and [1] give appropriate isomorphisms in this setting. Generalizing the definable cohomological machinery to coincide with the Galois cohomology measuring difference algebraic torsors defined by Bachmayr and Wibmer is also left to future work.

References

[1] Annette Bachmayr and Michael Wibmer. Trivializing torsors for difference algebraic groups. *Canadian Journal of Mathematics*, page 1–46, 2026.

[2] Zoé Chatzidakis and Anand Pillay. Generalized Picard–Vessiot extensions and differential Galois cohomology. *Annales de la Faculté des sciences de Toulouse : Mathématiques*, 2017.

[3] Pierre Deligne and James S. Milne. Tannakian categories, February 2022. Updated and corrected version of the article published in *Hodge Cycles, Motives, and Shimura Varieties*, Lecture Notes in Mathematics, vol. 900, pp. 101–228, 1982.

[4] Ehud Hrushovski. Computing the Galois group of a linear differential equation. *Banach Center Publications*, 58(1):97–138, 2002.

[5] Moshe Kamensky. Model theory and the tannakian formalism. *Transactions of the American Mathematical Society*, 367(2):1095–1120, 2015.

[6] Moshe Kamensky and Anand Pillay. Interpretations and differential Galois extensions. *International Mathematics Research Notices*, 2016(24):7390–7413, 04 2016.

[7] Ellis R. Kolchin. *Differential Algebraic Groups*. Academic press, 1985.

[8] Omar León Sánchez, David Meretzky, and Anand Pillay. More on Galois cohomology, definability, and differential algebraic groups. *The Journal of Symbolic Logic*, 89(2):496–515, 2024.

[9] Omar León Sánchez and Joel Nagloo. On parameterized differential Galois extensions. *Journal of Pure and Applied Algebra*, 220(7):2549–2563, 2016.

[10] Andy R. Magid. The complete Picard Vessiot closure. 2022. arXiv:2203.00705.

[11] A.R. Magid. *Lectures on Differential Galois Theory*. University lecture series. American Mathematical Society, 1994.

[12] David Meretzky. *Differential Field Arithmetic*. Phd thesis, University of Notre Dame, 2025. doi.org/10.7274/28786154.v1.

[13] David Meretzky. The short exact sequence in definable Galois cohomology. *The Journal of Symbolic Logic*, page 1–15, 2025. doi:10.1017/jsl.2025.8.

[14] David Meretzky and Anand Pillay. Picard-vessiot extensions, linear differential algebraic groups, and their torsors. *Communications in Algebra*, 0(0):1–14, 2024.

[15] David Meretzky and Anand Pillay. Galois theory, automorphism groups of prime models, and the Picard-Vessiot closure. 2026. Preprint. arXiv:2601.04076. math.LO.

[16] Andrei Minchenko and Alexey Ovchinnikov. Triviality of differential Galois cohomology of linear differential algebraic groups. *Communications in Algebra*, 47(12):5094–5100, 2019.

[17] A. Pillay and Ž. Sokolović. Superstable differential fields. *The Journal of Symbolic Logic*, 57(1):97–108, 1992.

[18] Anand Pillay. Remarks on Galois cohomology and definability. *Journal of Symbolic Logic*, 62(2):487–492, 1997.

[19] Anand Pillay. Differential Galois Theory I. *Illinois Journal of Mathematics*, 42(4):678 – 699, 1998.

[20] Anand Pillay. Algebraic D-groups and differential Galois theory. *Pacific Journal of Mathematics*, 216(2):343–360, 2004.

[21] Anand Pillay. The Picard–Vessiot theory, constrained cohomology, and linear differential algebraic groups. *Journal de Mathématiques Pures et Appliquées*, 108(6):809–817, 2017.

[22] Bruno Poizat. Modeles premier d'une theorie totalement transcendanty. *Groupe d'etude de theories stables*, tome 2, (1978 - 79).

[23] Bruno Poizat. Une théorie de galois imaginaire. *The Journal of Symbolic Logic*, 48(4):1151–1170, 1983.
[24] Omar León Sánchez and Anand Pillay. Differential Galois cohomology and parameterized Picard–Vessiot extensions. *Communications in Contemporary Mathematics*, 23(08):2050081, 2021.
[25] Jean-Pierre Serre. *Galois Cohomology*. Springer New York, New York, NY, 1979.
[26] Omar León Sánchez and David Meretzky. Parameterized D-torsors in differential Galois theory. *Preprint. ArXiv:2603.03658. math.LO*, 2026.
[27] Omar León Sánchez and Anand Pillay. Some definable Galois theory and examples. *The Bulletin of Symbolic Logic*, 23(2):145–159, 2017.
[28] Marius van der Put and Michael F. Singer. *Galois Theory of Linear Differential Equations*, volume 166 of *Graduate Texts in Mathematics*. Springer, 2003.

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